React – Fundamentals

Session - 3

1. Create a functional component named ProductCard that accepts props for productName and price, and displays them in a styled div.

Ans.

ProductCard.js:

```
import React from "react";

function ProductCard({ productName, price }) {
  return (
    <div
      style={{
        border: "1px solid #ccc",
        padding: "15px",
        width: "200px",
        borderRadius: "8px",
        textAlign: "center",
      }}
    >
      <h2>{productName}</h2>
      <p>Price: ₹{price}</p>
    </div>
  );
}

export default ProductCard;
```

App.js:

```
import React from "react";
import ProductCard from "./ProductCard";

function App() {
  return (
    <div>
      <ProductCard productName="Wireless Mouse" price={799} />
    </div>
  );
}

export default App;
```

2. Build a UserProfile component that takes username, followers, and profilePic as props and renders a mini Instagram-style profile card.

Ans.

UserProfile.js:

```
import React from "react";

function UserProfile({ username, followers, profilePic }) {
  return (
    <div
      style={{
        border: "1px solid #ddd",
        width: "250px",
        padding: "15px",
        borderRadius: "10px",
        textAlign: "center",
      }}
    >
      <img
        src={profilePic}
        alt="Profile"
        width="120"
        height="120"
        style={{ borderRadius: "50%" }}
      />

      <h2>{username}</h2>
      <p>{followers} Followers</p>
    </div>
  );
}

export default UserProfile;
```

App.js:

```
import React from "react";
import UserProfile from "./UserProfile";

function App() {
  return (
    <div>
      <UserProfile
        username="John"
        followers={1200}
        profilePic="https://via.placeholder.com/120"
      />
```

```
    </div>
  );
}

export default App;
```

3. Set up default props for your UserProfile component so that if followers or profilePic is not provided, it shows 0 followers and a default image URL.

Hint: Use the defaultProps property to set fallback values.

Ans.

UserProfile.js:

```
import React from "react";

function UserProfile({ username, followers, profilePic }) {
  return (
    <div
      style={{
        border: "1px solid #ddd",
        width: "250px",
        padding: "15px",
        borderRadius: "10px",
        textAlign: "center",
      }}
    >
      <img
        src={profilePic}
        alt="Profile"
        width="120"
        height="120"
        style={{ borderRadius: "50%" }}
      />

      <h2>{username}</h2>
      <p>{followers} Followers</p>
    </div>
  );
}

UserProfile.defaultProps = {
  followers: 0,
  profilePic: "https://via.placeholder.com/120",
};

export default UserProfile;
```

App.js:

```
import React from "react";
import UserProfile from "./UserProfile";

function App() {
  return (
    <div>
      <UserProfile username="John" />
    </div>
  );
}

export default App;
```

4. Add prop type validation to your ProductCard component using the prop-types package to ensure productName is a string and price is a number.

Hint: Install prop-types with npm and import it in your file.

Ans.

Install prop-types:

```
npm install prop-types
```

ProductCard.js:

```
import React from "react";
import PropTypes from "prop-types";

function ProductCard({ productName, price }) {
  return (
    <div
      style={{
        border: "1px solid #ccc",
        padding: "15px",
        width: "200px",
        borderRadius: "8px",
        textAlign: "center",
      }}
    >
      <h2>{productName}</h2>
      <p>Price: ₹{price}</p>
    </div>
  );
}

ProductCard.propTypes = {
```

```
  productName: PropTypes.string.isRequired,
  price: PropTypes.number.isRequired,
};

export default ProductCard;
```

App.js:

```
import React from "react";
import ProductCard from "./ProductCard";

function App() {
  return (
    <div>
      <ProductCard productName="Laptop" price={55000} />
    </div>
  );
}

export default App;
```